Journal and Proceedings, Royal Society of N.S.W., Vol. 94, pp. 227–232, 1961

The Palaeomagnetism of some Igneous Rock Bodies in New South Wales

R. Boesen, E. Irving and W. A. Robertson *

(*Received April* 5, 1960)

Introduction

The directions and intensities of magnetization of rock specimens from four igneous bodies in eastern New South Wales have been measured. These are labelled A, B, C and D in Figure 1. To ensure freshness samples were taken from as far below the natural surface as practicable, all specimens being taken from road cuttings and quarries. The bodies studied do not appear to have suffered any appreciably post-consolidation tilting. The collecting sites are located by map references which are taken from the one-inch-to-the-mile military series and are given by the sheet title followed by the six-figure grid reference. The results indicate a Mesozoic age for these bodies which is older than had been previously supposed.

A. *Prospect Intrusion*—This is a dish-shaped sheet of teschenitic dolerite convex downwards, intruded discordantly (at least in part) into the upper beds of the Liverpool Subgroup (Lovering, 1954) of the Wianamatta Group, which is of Upper Triassic age. It has a chilled envelope enclosing a main mass which is variable in texture and composition. David (1950, p. 578) states that on petrological grounds this dolerite, among other basic hypabyssal intrusives in eastern Australia, is usually regarded as Tertiary. Samples taken from three quarries (Liverpool 920215, 914205, 916216), included specimens from the chilled envelope and the more pegmatitic facies.

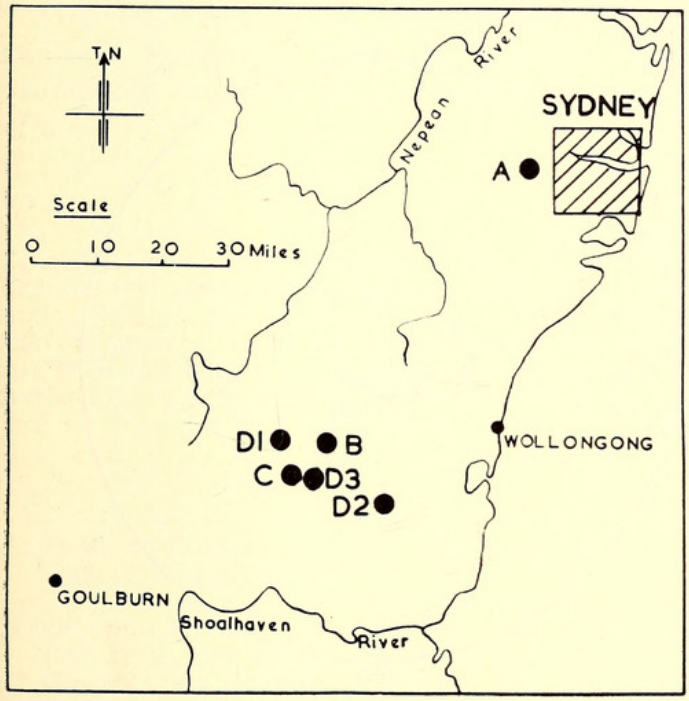


FIG. 1

Locality sketch-map. The localities are labelled as follows: A, Prospect Dolerite; B, Gibraltar Syenite; C, Gingenbullen Dolerite; D, Tertiary Basalts (D1 Berrima, D2 Robertson, D3 Moss Vale)

B. *Gibraltar Syenite*—This is thought to be an asymmetric laccolith and is intruded into Hawkesbury Sandstone (Stevens, 1956). It is composed of an aegerine-augite microsyenite with deuterically altered phases and narrow pegmatitic veins. On petrological grounds it is regarded as probably early Tertiary (David, p. 581; Stevens, 1956). The samples were obtained from the two operating quarries on the southern face (Mittagong 426426, 427424). Another disused quarry a little east of these (Mittagong 431421) and much nearer the roof showed signs of weathering and was not sampled.

C. *Gingenbullen Dolerite*—This is a 350 ft thick horizontal tabular body composed of columnar dolerite. The mass is either a denuded sill or a dyke. It lies on the uppermost beds of the Liverpool Subgroup and is thus of post-Upper Triassic age. Samples were taken at various heights in the exposed face of the disused quarry on the northern side of the body (Moss Vale 317347).

D. *Some Tertiary Basalts*—These have been sampled from the Berrima, Moss Vale and Robertson areas. At Berrima the basalt occurs as a horizontal sheet overlying shales of the lower part of the Liverpool Subgroup. It probably represents a residual of a now extensively dissected flow. Fresh rock was found in only one road cutting which was sampled at regular intervals over a distance of about fifty yards (Mittagong 370473). The rock is a

* Department of Geophysics, Australian National University, Canberra, A.C.T.

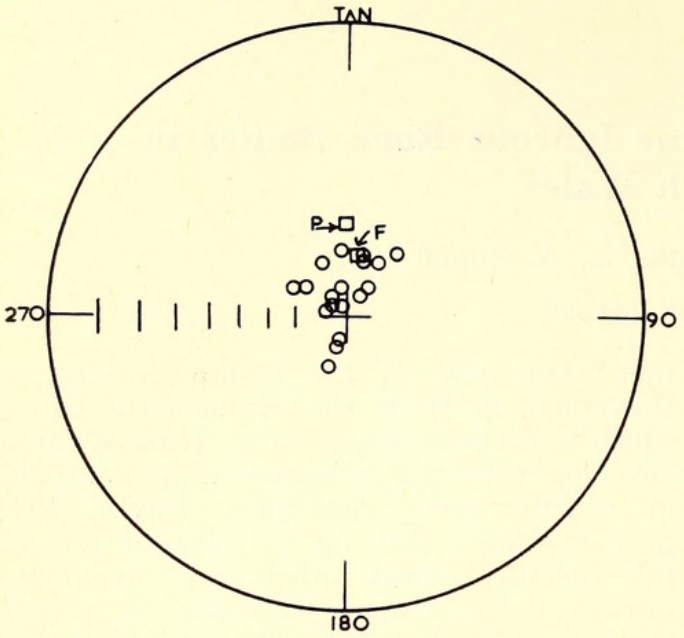


FIG. 2
Directions of NRM in the Prospect Dolerite
North-seeking directions of magnetization are plotted as circles on the upper hemisphere. In this, and all subsequent figures, polar stereographic projections are used, and the dipole (**P**) and present (**F**) field directions are marked

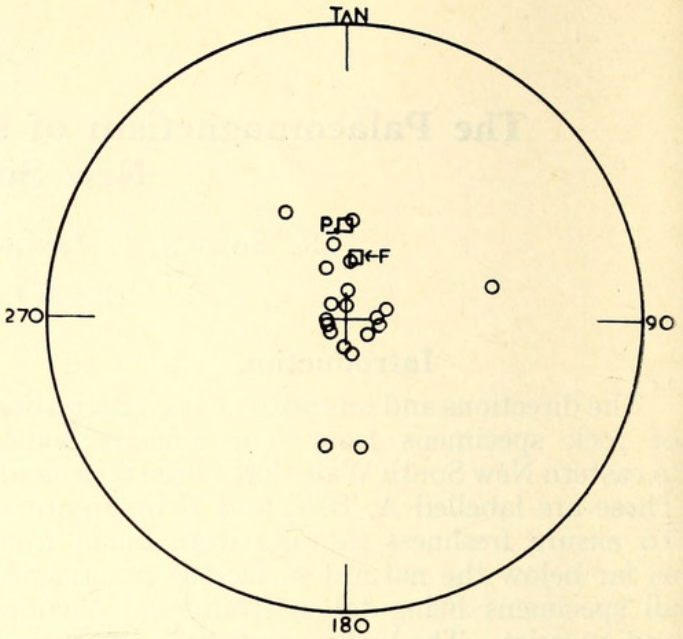
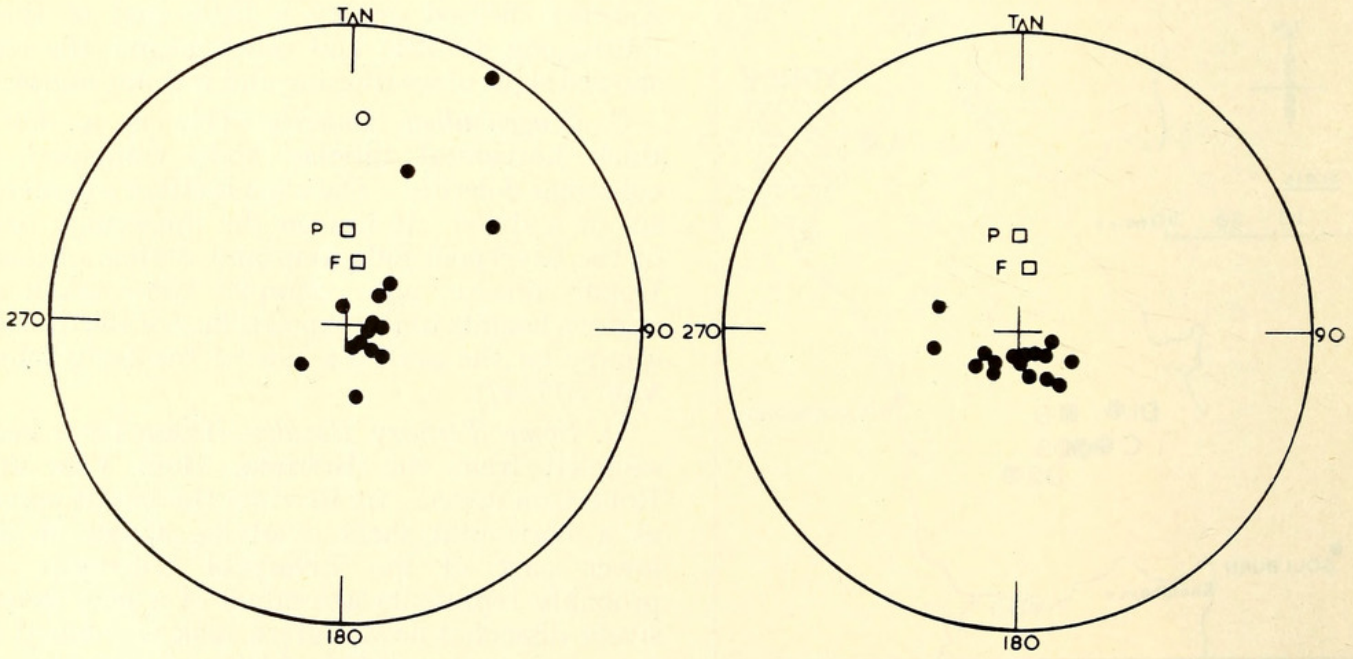


FIG. 3
Directions of NRM in the Gibraltar Syenite
North-seeking directions are plotted as circles on the upper hemisphere

porphyritic olivine basalt, variable in texture and containing phenocrysts of olivine in a groundmass of microlitic plagioclase, augite, magnetite, and occasional glass. The basalt of the Robertson flow is petrographically very similar to the Berrima rock. Samples were taken from road cuttings (Kiama 627294, 594275, 560284). The weathering in most exposures was severe but large unweathered nodules could be obtained in some exposures. The samples from Moss Vale were taken from a disused quarry (Moss Vale 364337). In addition to these localities



(*a*) (*b*)

FIG. 4
Directions of magnetization in the Gingenbullen Dolerite
North-seeking directions are plotted as circles on the upper and dots on the lower hemisphere. (*a*) NRM, the spread of points towards the dipole field (**P**), is due to partial instability in some specimens. (*b*) Directions after treatment in an alternating peak field of 150 oersteds which removes this instability

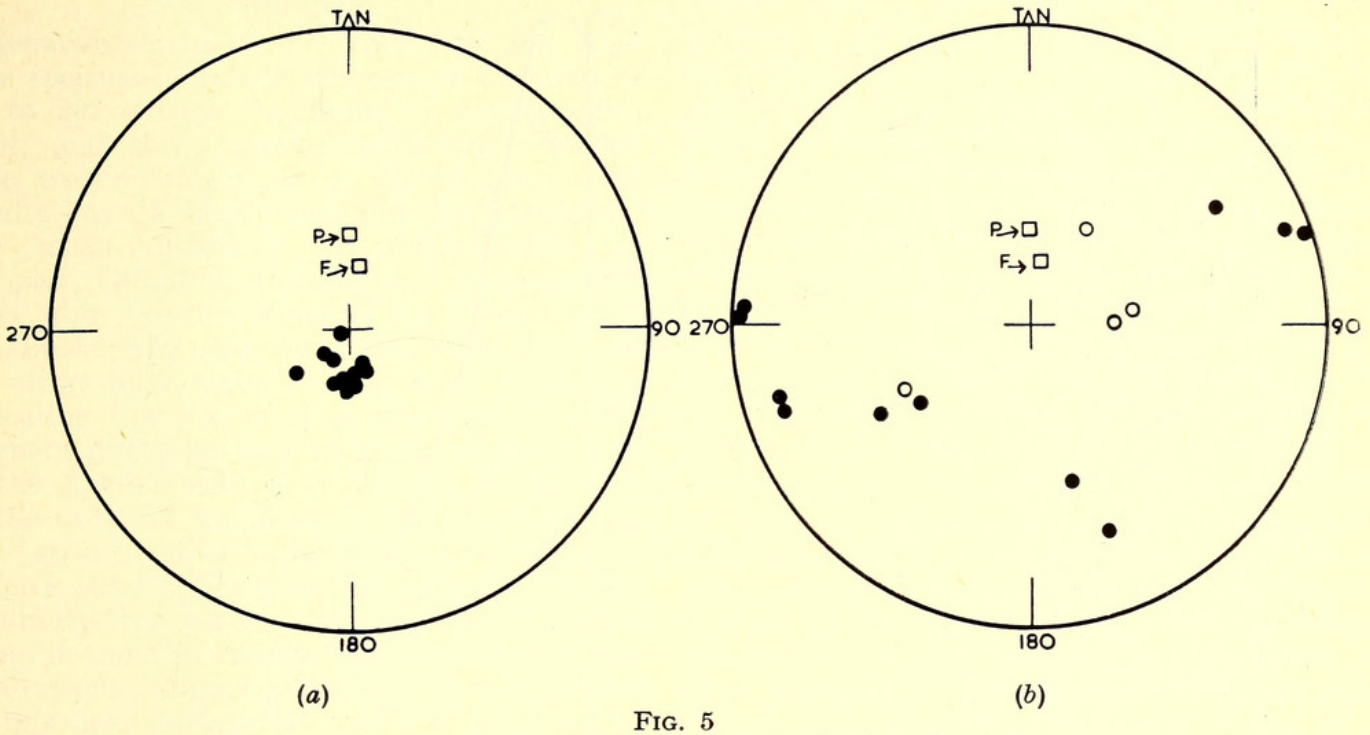


FIG. 5
Direction of magnetization in some Tertiary basalts
(*a*) Berrima basalt, stable magnetization. (*b*) Robertson and Moss Vale flows, randomly directed unstable magnetization

other basalts in the Robertson-Exeter-Mittagong district were examined, but were found to be extensively weathered. Similarly on the Illawarra scarp and adjoining highlands exposures of the Cordeaux teschenitic dolerite, the Bong Bong basalt on and near Saddleback Mountain (Kiama 817516 west to 787163, and 783155 west to 773159), and the whole of the exposed Kangaroo Mountain basanite, were visited but found to be too decomposed to be of use.

Remanent Magnetization

The hand specimens were oriented prior to extraction from the rock face. Cylinders were machined from these with non-magnetic tools. The directions of magnetization were measured using an astatic magnetometer. The experimental error involved throughout all these operations is about 5° in the determined direction.

The directions of natural remanent magnetization (NRM) are plotted in Figures 2 to 5 on

TABLE 1
Mean Directions of N.R.M.

The values given refer to the four igneous occurrences listed in the first column. In the case of the Gingenbullen Dolerite the values obtained after removal of the unstable magnetization by treatment in 150 oersteds peak alternating field are also given. S is the number of oriented rock samples, N is the number of specimens cut from these samples, D and I are the declination and inclination of the mean direction, R is the resultant giving each specimen unit weight, α is the 95% error in the mean direction (Fisher, 1953) and k is Fisher's precision parameter. ΔP and ΔF are the angular distances between the mean directions and the dipole and present field respectively

	S	N	D	I	R	α ($P=0{\cdot}05$)	k	ΔP	ΔF	Pole Positions
A. Prospect Dolerite (33° 49′ S, 150° 49′ E)	10	18	359	—81	17·40	6·8	28	27	16	51 S, 151 E
B. Gibraltar Syenite (34° 28′ S, 150° 26′ E)	10	20	27	—86	17·94	11·7	9	32	22	41 S, 146 E
C. Gingenbullen Dolerite (34° 22′ S, 150° 20′ E)	8	16								
1. N.R.M.			56	+71	12·89	19·6	5	147	165	— —
2. After treatment			191	+80	15·33	8·0	23	154	166	53 S, 144 E
D1. Tertiary Basalt, Berrima	7	11	187	+76	10·78	6·9	44	158	169	— —

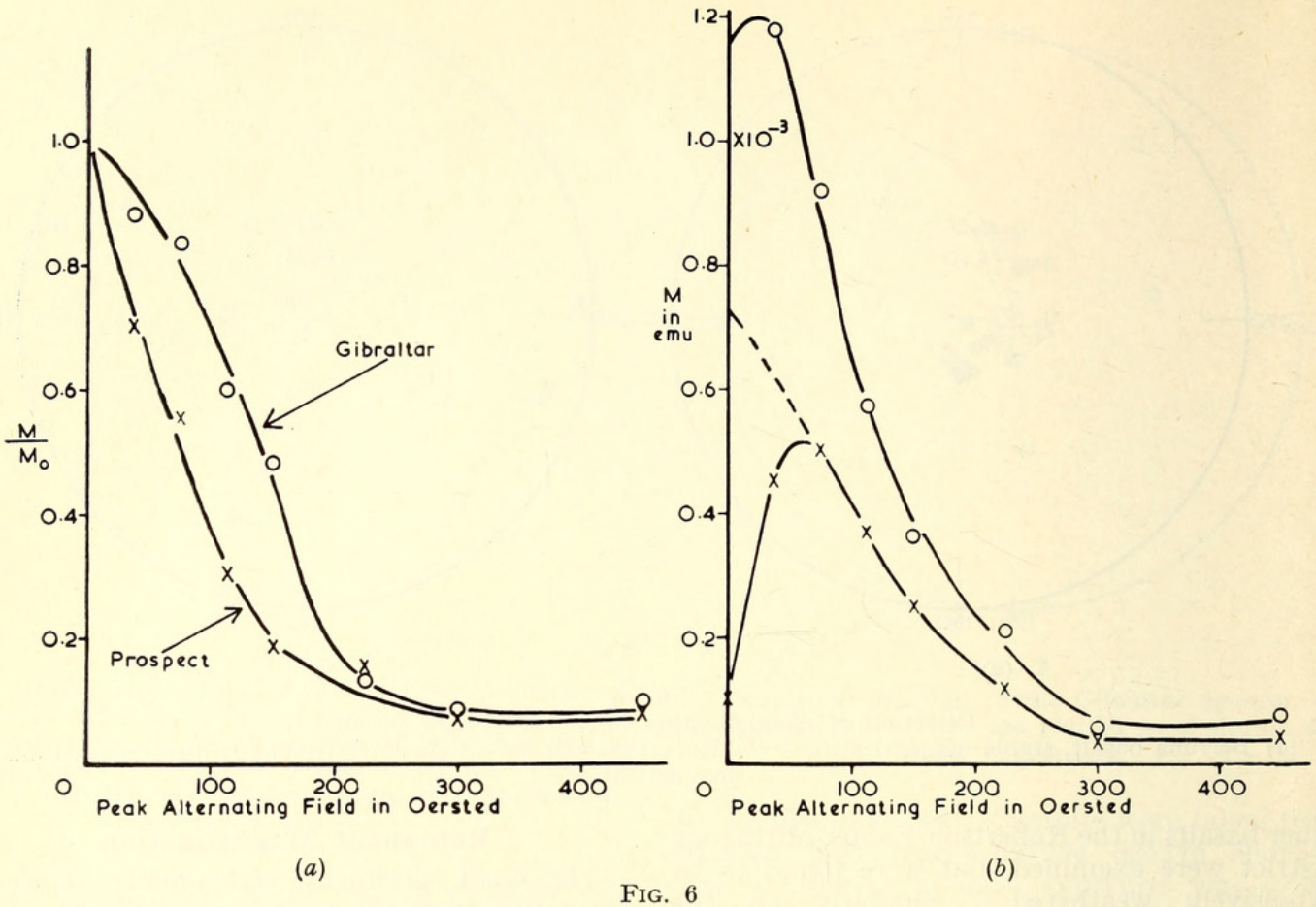


FIG. 6

Alternating field demagnetization

(*a*) Prospect Dolerite and Gibraltar Syenite specimens, M/M_0 is plotted against peak alternating field; M is the intensity of magnetization after treatment and M_0 is the intensity of NRM.

(*b*) Gingenbullen Dolerite, M is plotted against alternating field. The upper curve is from a specimen with a predominantly stable magnetization and the lower curve is from a specimen in which the stable and unstable components are of comparable magnitudes.

stereographic projections, the convention adopted being to plot the north-seeking direction of magnetization as circles on the upper hemisphere and as dots on the lower hemisphere. The mean directions are given in Table 1. The directions in the Prospect Dolerite (Fig. 2) and Gibraltar Syenite (Fig. 3) are well grouped, the scatter being greater in the latter. Both have steep negative inclinations. The directions at Gingenbullen are for the most part almost vertical with positive inclination. There are, however, a few results with northerly declinations and less steep inclination which are strung out towards the geocentric axial dipole field. The directions in the Berrima basalt show a close grouping with steep positive inclinations. The directions of magnetization at Robertson and Moss Vale (Fig. 5*b*) show a wide scatter, the resultant R for 15 specimens being 1·32 and from the tables given by Watson (1956) the distribution is random at $P=0{\cdot}05$.

At Prospect the intensities of natural remanent magnetization (in emu/cc $\times 10^{-3}$) range from 1 to 8, at Gibraltar from 0·1 to 1·0, at Gingenbullen from 0·1 to 1·2 and in the Tertirya Basalts from 1–10.

The saturation isothermal remanence ($M_{i(\text{sat})}$) and the field required to saturate ($H_{(\text{sat})}$) and the coercivity (H_c) of $M_{i(\text{sat})}$ for specimens from the three intrusives are given in Table 2.

TABLE 2

Magnetic Properties

M is the intensity of the natural remanent magnetization, $M_{i(\text{sat})}$ is the isothermal remanent magnetization both in emu/cc $\times 10^{-3}$. $H_{(\text{sat})}$ is the field required for saturation and H_c is the back field needed to remove $M_{i(\text{sat})}$ in oersted

	M	$M_{i(\text{sat})}$	$H_{(\text{sat})}$	H_c
A. Prospect Dolerite	6·3	263	1500	200
B. Gibraltar Syenite	0·5	208	2000	250
C. Gingenbullen Dolerite	0·1	62	1000	200

Stability

Remeasurement of the Robertson and Moss Vale specimens showed changes of direction of up to 20° outside the limits of experimental error, and this, together with the observed wide scatter of directions, shows that these basalts have a highly unstable magnetization. The mean directions of magnetization at Prospect, Gibraltar, Berrima and Gingenbullen differ from the directions of the present (ΔF) and dipole (ΔP) fields by amounts which exceed the errors by a factor of 2 or more (Table 1), indicating that for most of the specimens the magnetic directions have remained little changed in the Earth's field (0·6 oersted) for periods of the order of hundreds of years. However, the " strung " distribution observed at Gingenbullen is of a familiar type associated with partial instability, the vertical reversed magnetization being, in some specimens, substantially affected by a viscous component imposed by the present Earth's field during the past few hundreds of years.

Tests have been made by treating specimens in alternating magnetic fields in the absence of any steady field using the apparatus and methods described by Irving, Stott and Ward (1961). In fields up to 300 oersteds (peak) the effect in direction on specimens from Prospect is negligible. The effect in Gibraltar specimens is negligible up to 200 oersteds. All specimens in these two intrusions were treated in alternating fields of 150 oersteds with no important effects on the two distributions. The effect on intensity is illustrated in Figure 6.

In the specimens from Gingenbullen the viscous components due to the present earth's field are removed in fields of about 100 oersteds. All specimens have been treated at 150 oersteds and the results are plotted in Figure 4*b*. The associated changes of intensity are given in Figure 6*b* for two specimens ; one in which the initial direction is vertical and stable, being little changed in direction in fields up to 300 oersteds, and a second in which the initial direction has only a shallow dip, and is substantially affected by an unstable component. In the first case (upper curve) there is a very small increase of intensity due to the removal of a small unstable component, and in the second case the removal of the much larger unstable component causes an increase of intensity by a factor of four after treatment in low alternating fields.

The magnitudes S and U of the stable and unstable components **S** and **U** in this latter case may now be estimated. The NRM has a direction of (52, +18) and a magnitude 114×10^{-6} emu/cc. and is the vector sum **S**+**U**. **U** is directed along the dipole field (0, −54) and the direction of **S** (202, +75) is obtained after **U** is removed by treatment in low alternating fields ; in this case 75 oersteds was used. (The intensity measured after this treatment does not, of course, give S directly, since the latter is somewhat diminished by this treatment). The value obtained for S is 722×10^{-6} (see extrapolated dotted curve Fig. 6*b*) and for U is 727×10^{-6} emu/cc. It may be noted that S and U have each about half the magnitude of the NRM (1150×10^{-6}) of the stable specimen (upper curve in Fig. 6*b*).

Although the demagnetization characteristics of these specimens do not show the high stability which is associated with the NRM of some igneous rocks the results do, nevertheless, conform to the initial palaeomagnetic condition for stability, namely, that the directions, in most cases, have been little affected by a component due to the earth's field in recent times, and in those cases where a substantial unstable component is present its effect may be removed by treatment in low alternating fields.

The stability of the basalts of the Berrima, Moss Vale and Robertson areas has been studied as part of a general palaeomagnetic study of the Tertiary Basalts of New South Wales.



FIG. 7

Pole Positions

The pole positions for the three intrusions studied are labelled as in Table 1. The Cenozoic and Mesozoic poles previously obtained from Australia are numbered as follows : 1, Newer Volcanics of Victoria (Upper Pliocene, to Recent) ; 2, Older Volcanics of Victoria (Lower Tertiary) (Irving and Green 1958) ; 3, Tasmanian Dolerite Sills (Late Triassic, Jurassic or Cretaceous) (Irving 1956)

The results, which have been described by Irving, Stott and Ward (1961), are not discussed further in this paper.

Discussion

The directions of NRM in the Prospect Dolerite, the Gibraltar Syenite, and the directions in the Gingenbullen Dolerite after treatment in 150 oersteds are stable, and may be identified with the direction of the geomagnetic field at the time these intrusions cooled. On this assumption, and further assuming that the geomagnetic field at these times was on average that of a geocentric dipole, the pole positions (south) consistent with these directions may be calculated (Table 1). These poles are the points at which this geocentric dipole intersected the earth's surface. It has to be remembered that errors are present in these determinations since the time span of several hundreds of years necessary to average out the secular variation of the earth's magnetic field is not certainly represented in each case. In the larger bodies (Prospect and Gibraltar) this error is possibly quite small since these bodies may have taken a long time to cool.

The pole positions for the three intrusive bodies are plotted in Figure 7. They are in the region of Tasmania near the pole obtained for the Tasmanian dolerites (pole 3) but in a lower latitude than that obtained from the Older Volcanics of Victoria (pole 2). This suggests that these intrusions are older than the latter, and of an age comparable to the former, and are therefore Mesozoic.

References

David, T. W. E., 1950. Geology of the Commonwealth of Australia. Arnold, London.

Fisher, R. A., 1953. Dispersion on a Sphere. *Proc. Roy. Soc.*, A, **217**, 295.

Irving, E., 1956. Magnetization of the Tasmanian Dolerites. *Pap. and Proc. Roy. Soc. Tasmania*, **90**, 157.

Irving, E., and Green, R., 1958. Polar Movement relative to Australia. *Geophys. J. Roy. Astr. Soc.*, **1**, 64.

Irving, E., Stott, P. M., and Ward, M. A., 1961. Demagnetization of Igneous Rocks by Alternating Magnetic Fields. *Phil. Mag.* (in press).

Lovering, J. F. 1954. The stratigraphy of the Wiannamatta, Group, Triassic System, Sydney Basin. *Rec. Aust. Museum*, **23**, 169.

Stevens, R D. 1956. Observations on Mount Gibraltar. *J. Proc. Roy. Soc. N.S.W.*, **90**, 100.

Watson, G. S., 1956. A Test for Randomness of Directions. *Mon. Not. Roy. Ast. Soc., Geophys. Supp.* **7**, 160.



Irving, E, Boesen, R, and Robertson, W A. 1961. "The palaeomagnetism of some igneous rock bodies in New South Wales." *Journal and proceedings of the Royal Society of New South Wales* 94(6), 227–232. https://doi.org/10.5962/p.360794.

View This Item Online: https://www.biodiversitylibrary.org/item/174326
DOI: https://doi.org/10.5962/p.360794
Permalink: https://www.biodiversitylibrary.org/partpdf/360794

Holding Institution
Smithsonian Libraries and Archives

Sponsored by
Biodiversity Heritage Library

Copyright & Reuse
Copyright Status: In Copyright. Digitized with the permission of the rights holder
Rights Holder: Royal Society of New South Wales
License: http://creativecommons.org/licenses/by-nc-sa/3.0/
Rights: https://www.biodiversitylibrary.org/permissions/

This document was created from content at the **Biodiversity Heritage Library**, the world's largest open access digital library for biodiversity literature and archives. Visit BHL at https://www.biodiversitylibrary.org.

This file was generated 29 July 2025 at 01:58 UTC